PHASE III • CODE TO CONNECT

RiskPulse AI

Explainable Financial Risk Intelligence

Turn public financial narratives into traceable risk signals,
constraint-aware index analysis and synthetic multi-asset stress.

PUBLIC EVIDENCE
Marketaux + Bluesky

→

RISK CONTEXT
Sentiment + events + impact

→

ANALYST LABS
Index rebalance + stress

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Research prototype • Public and synthetic data • Analyst-reviewed decisions

Problem & Approach

Three gaps between financial news and a reviewable risk decision

01 FRAGMENTED INFORMATION	**TRACEABLE SOURCE LAYER**
News and social posts have different formats and timing. Repeated headlines obscure the originating event.	Marketaux + Bluesky → normalize, ticker-match and deduplicate; keep timestamps and source links in SQLite.
02 INCOMPLETE RISK CONTEXT	**THREE COMPLEMENTARY AI SIGNALS**
Headline polarity cannot distinguish macro, credit or corporate events—or quantify their likely importance.	FinBERT sentiment [−1,+1] + five event probabilities + XGBoost reaction-magnitude decile [1–10].
03 SIGNAL-TO-DECISION GAP	**TWO REVIEWABLE FINANCIAL LABS**
A risk score alone does not show which portfolio shift is feasible or what a hypothetical shock would affect.	15-stock constrained Index Lab + 30-position synthetic Stress Lab, with explicit assumptions and controls.

Beyond sentiment-only triage: evidence → event context → bounded simulations; no automatic trading.

System Design & Data Flow

A source-linked risk pipeline that branches into two analyst applications

PUBLIC TEXT	INGEST + STORE	FINANCIAL NLP	IMPACT MODEL	DELIVERY
Marketaux news Bluesky posts	Normalize • ticker-match Deduplicate • SQLite	FinBERT sentiment 5-class event model	XGBoost severity decile	FastAPI JSON React Quick / Analyst

HISTORICAL CONTEXT → pre-publication price/volume + sentiment + event probabilities → XGBoost impact estimate

FINANCIAL APPLICATIONS

MODULE A / INDEX LAB	MODULE B / STRESS LAB
15 equal-weight reference equities SciPy SLSQP constrained rebalancing Stock ±2 pp • Sector ±5 pp • Turnover ≤10%	30 synthetic positions: 12 loans + 10 bonds + 4 swaps + 4 options Instantaneous MTM ≠ incremental 12-month ECL

RESEARCH INPUTS: 5-class public event corpus • FiQA sentiment • FNSPID news / stock prices

Optional Bujji explanations require Groq configuration; archived and demo data are separate from live NLP inference.

Severity estimates reaction magnitude—not direction. Stress inputs are hypothetical, not forecasts.

Implementation Highlights

Modules, technology choices and why each component matters

01 TRACEABLE INGESTION

TECH Python adapters • SQLite

WHY Source URLs, timestamps, ticker matches and deduplication let reviewers inspect each signal.

DELIVERS Auditable normalized event records.

02 FINANCIAL RISK ENGINE

TECH FinBERT regression + 5-class head; XGBoost

WHY Finance-aware NLP + 17 impact inputs (14 numeric / 3 categorical), including past-only prices.

EVENTS Geopolitical • Macroeconomic • Credit Event
Merger/Acquisition • Product Launch

03 CONSTRAINT-AWARE INDEX LAB

TECH SciPy SLSQP • Pandas • 15 long-only equities

WHY Rebalancing must satisfy full investment and enforceable position, sector and turnover limits.

CONTROLS ±2 pp stock • ±5 pp sector • ≤10% one-way turnover • 10 bps costs

04 SYNTHETIC MULTI-ASSET STRESS

TECH Python pricing • NumPy • Pydantic

WHY Different instruments react differently; loan ECL and trading-book MTM have different horizons.

METHOD 12 loans • 10 bonds • 4 swaps • 4 options
What-if scaling + sampled-grid reverse stress

PLATFORM Python / PyTorch / Transformers / XGBoost • FastAPI structured JSON • React + TypeScript/Vite

Models output financial sentiment [−1,+1], five event probabilities, and a severity decile [1–10].

Key Results & Recorded Outputs

Evidence saved in training notebooks and repository JSON — not a new live rerun

99.46%
EVENT ACCURACY
1,658 real-only test examples

0.9798
EVENT MACRO-F1
Same archived classifier experiment

16,576
CLEANED EVENTS
13,260 train / 1,658 val / 1,658 test

MODULE A / SAVED REBALANCE

AAPL allocation in bundled example

6.67% → **7.09%**

Before → after | +0.42 percentage points

0.42% one-way turnover • 357-session inputs

MODULE B / SAVED SEVERE CREDIT STRESS

INSTANTANEOUS MTM P&L	INCREMENTAL 12-MONTH ECL
−$269.5k	**+$27.7k**

Different time horizons; not additive portfolio NAV.

WHAT CHANGES VS A NAÏVE SENTIMENT-ONLY ALERT?

Polarity alone → limited context | RiskPulse → source trace + event category + impact magnitude + bounded simulations

Evidence caveat: Credit Event test support = 12; no quantified end-to-end speedup or live investment return is claimed.

Recorded results only: missing fine-tuned NLP checkpoints prevent clean-clone inference reproduction.

Domain Impact: The Analyst Workflow

From source inspection to bounded allocation and transparent stress discussion

01 TRIAGE

Which event deserves review?

Source link + ticker + time
Financial sentiment + five-class event
Impact severity / ranking context

More structured evidence review

02 ALLOCATE

Which weight change is feasible?

Long-only 15-stock benchmark
Stock, sector and turnover limits
Inspect proposed weights and costs

Control-aware portfolio proposals

03 STRESS

What happens under a shock?

Scenario assumptions are explicit
Loans / bonds / swaps / options
Instrument attribution + distinct horizons

Explainable sensitivity discussions

HUMAN IN THE LOOP • Inspect sources • Review constraints • Approve hypothetical scenarios

Research and analyst exploration only; no autonomous trading, client data or regulatory-capital claims.

Limitations & Next Steps

Distinguish the demonstrated prototype from unverified full-model reproducibility

SUPPLIED / DEMONSTRATED

- Source-linked ingestion and saved records
- 15-stock Index Lab and 30-position Stress Lab
- Archived model experiments and example outputs
- Prior test logs: 120 backend + 14 frontend passed

OPEN LIMITATIONS

- Two original fine-tuned NLP weight sets absent
- Some original training splits / price archives external
- Event-training initializer ≠ runtime base config
- Cloud preview uses saved demo fixtures, not full NLP inference

NEXT VALIDATION MILESTONES

01 PUBLISH ARTIFACTS

Authentic checkpoints + hashes; dataset provenance and redistribution rights.

02 VERIFY CLEAN CLONE

Run E2E inference with genuine weights; independently verify demo behavior.

03 TEST GENERALIZATION

Prospective cross-source events, low-support classes, and regime sensitivity.

RiskPulse AI connects narrative evidence to interpretable, reviewable financial simulations.

Public / synthetic data • No live trades or guaranteed forecasts • Thank you